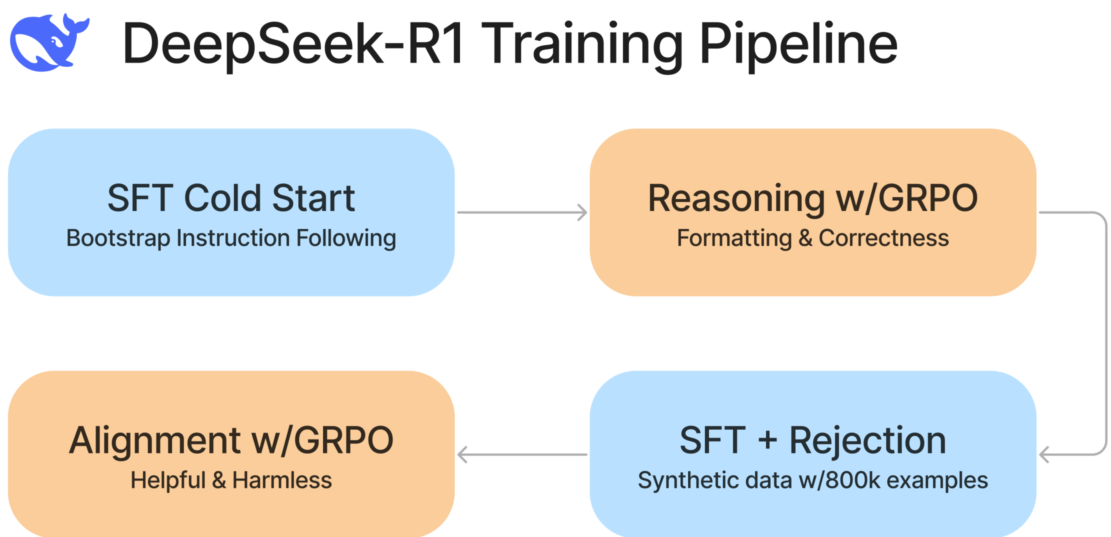
AIMS5740 Final Project

Topic 1 — Data Selection + RL for LLMs (Math / STEM)



1. Background

LLMs often underperform on math and STEM reasoning not due to lack of capacity, but because:

- training data is noisy or misaligned,
- SFT optimizes imitation rather than correctness,
- reasoning errors compound across steps.

Recent work shows that **data curation + RL-style post-training** can significantly improve reasoning accuracy.

2. Related Work

You are encouraged to cite and relate your project to:

- **MetaMath / MetaMathQA**

  https://meta-math.github.io/

  (large-scale math data curation and training signals)

- **GRPO (Group Relative Policy Optimization)**

  https://huggingface.co/learn/cookbook/en/fine_tuning_llm_grpo_trl

  (practical RL recipe for reasoning-oriented post-training)

- **Verifier-based RL for math reasoning**

  https://arxiv.org/abs/2110.14168

(training verifiers to guide math problem solving)

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3. Suggested Base Models (similar to hm1)

Choose **one**:

- `Qwen2.5-0.5B`
- `Llama-3.2-1B-Instruct`

Smaller models are preferred for RL due to stability and iteration speed.

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4. Pipeline Overview

Stage A — Data Selection

Choose at least one:

- https://huggingface.co/datasets/zwhe99/DeepMath-103K
- SynthLabsAI/Big-Math-RL-Verified
- STEM subset from instruction data (verifiable)

Apply **explicit filtering rules**, e.g.:

- remove ambiguous problems,
- enforce clean final-answer formats,
- length and difficulty control.

Stage B — Supervised Fine-Tuning (SFT)

- Framework: llama-factory
- Output: Base vs SFT model comparison (loss, evaluation accuracy)

Stage C — Reinforcement Learning

Choose:

- **Verifier-based reward** (exact match)

Reward design must include **≥2 components**, e.g.:

- correctness reward,
- format or reasoning-structure reward.

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5. Evaluation Benchmarks

Use `lm-eval-harness` or other standard evaluation pipeline, or the official evaluation code:

- HuggingFaceH4/MATH-500
- https://huggingface.co/datasets/openai/gsm8k
- https://huggingface.co/datasets/TIGER-Lab/TheoremQA

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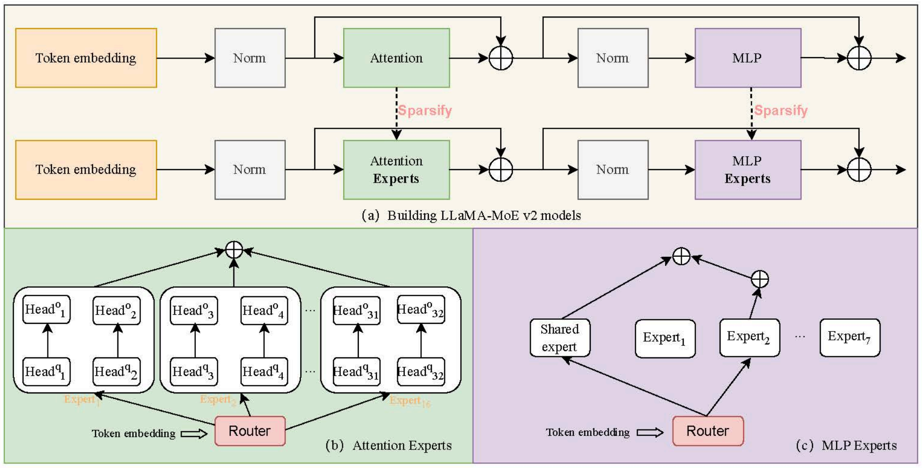
6. Analysis Expectations

- Does RL improve correctness but hurt general QA?
- Sensitivity to data filtering?

- Typical failure modes (format drift, reward hacking, general performance change).

Topic 2 — LLaMA-MoE (SFT Only)



1. Background

Mixture-of-Experts (MoE) models increase capacity by activating **only a subset of parameters per token**.

This project is based on:

LLaMA-MoE v2: Exploring Sparsity of LLaMA from Perspective of Mixture-of-Experts

https://arxiv.org/abs/2411.15708 or any other frameworks that enable MOE on LLM.

The key idea is **post-training sparsification** (no expensive continual pretraining).

2. Core References

- Paper: https://arxiv.org/abs/2411.15708
- Code: https://github.com/OpenSparseLLMs/LLaMA-MoE-v2

Related works:

- **Switch Transformer**https://arxiv.org/abs/2101.03961
- **LLaMA-MoE (continual pretraining variant)**https://arxiv.org/abs/2406.16554
- **Dynamic data mixing for MoE instruction tuning**https://arxiv.org/abs/2406.11256

3. Suggested Base Models

Similar to assignment 1 that can fit to the mem

4. Pipeline Overview

Step A — Dense → MoE Conversion

Choose:

- **MLP-MoE**
- **Attention-MoE**

Report:

- number of experts,
- top-k routing,
- load-balancing loss.

Step B — SFT-only Post-Training

Instruction data (examples):

- `SlimOrca`
- `OpenHermes`
- `LIMA`
- Optional math/code: `MetaMathQA`

Step C — Routing Analysis (Required)

Analyze:

- expert utilization,
- layer-wise routing patterns,
- specialization across datasets.

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5. Evaluation Benchmarks

Following LLaMA-MoE v2 or:

- MMLU (5-shot)
- GSM8K (8-shot)
- ARC-Challenge (25-shot)
- HellaSwag (10-shot)
- Winogrande (5-shot)
- PIQA (0-shot)
- BoolQ (paper uses 32-shot; 0-shot acceptable for course)

Or some easier benchmark:

- **GSM8K (0-shot or 2-shot)**
  - small input size, clear correctness signal
  - good for observing expert specialization on math reasoning
- **ARC-Easy (0-shot)**
  - much lighter than ARC-Challenge
  - suitable for small-capacity models

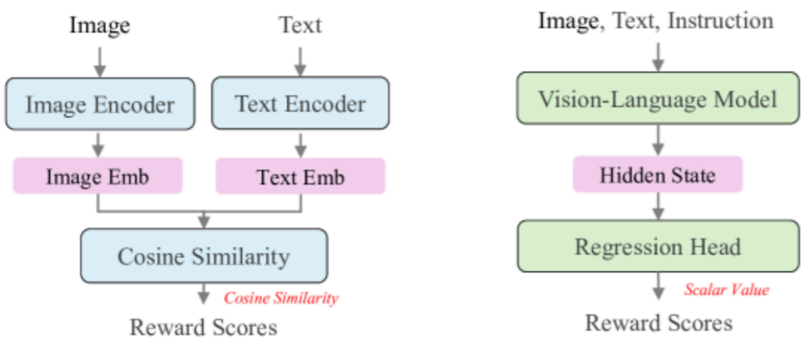
Evaluation Goal

The goal of evaluation is **not leaderboard performance**, but to analyze:

- performance gap between **dense vs MoE** models,
- trade-offs between **sparsity and accuracy**,
- how routing behavior correlates with task types.

Students are encouraged to **justify their benchmark choice** based on model size and compute constraints.

Topic 3 — Text-to-Image Generation with RL (Understanding-based Reward)



CLIP-based RM Architecture **VLM-based RM Architecture**

1. Background

Text-to-image models often:

- generate realistic images,
- but hallucinate objects, attributes, or spatial relations.

This project aligns generation using **image understanding as the reward**.

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2. Related Work & Benchmarks

In this project context, we focus on **recent advances in using reinforcement learning (RL) and multimodal large language models (MLLMs) as reward models** to improve generation quality, especially in text-to-image (T2I) and unified multimodal tasks.

T2I RL with Multimodal Rewards

- **ReasonGen-R1** – A unified multimodal reasoning framework that uses **GRPO (Group Relative Policy Optimization)** for reward-guided training of multimodal models.

  Repository: https://github.com/Franklin-Zhang0/ReasonGen-R1

  This is a concrete implementation of RL on a multimodal model (LLM + vision), illustrating how a **reward-based post-training strategy** can improve reasoning across modalities.

- **T2I-R1**.

  Repository: https://github.com/CaraJ7/T2I-R1/tree/main

Recent VLM-based Reward Modeling

Recent research has proposed using *large multimodal models* as reward evaluators for text-to-image and unified generation tasks:

- **Multimodal LLMs as Reward Models for T2I** (LLaVA-Reward, ICCV 2025)

This work designs a multimodal large language model to produce reward scores that jointly assess semantic alignment, image fidelity, and instruction adherence.

Paper: https://openaccess.thecvf.com/content/ICCV2025/papers/Zhou_Multimodal_LLMs_as_Customized_Reward_Models_for_to-Image_Generation_ICCV_2025_paper.pdf

Evaluation benchmarks:

- **T2I-CompBench (NeurIPS 2023)**https://proceedings.neurips.cc/paper_files/paper/2023/file/f8ad010cdd9143dbb0e9308c093aff24-Paper-Datasets_and_Benchmarks.pdf
- **TIFA**https://tifa-benchmark.github.io/
- **GenEval-2**https://arxiv.org/abs/2512.16853
- **GenAI-Bench**https://openreview.net/pdf?id=d1b464d7c538923ddbca638886afdc54d57ebae7

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3. Suggested Base Generators

Choose:

- https://huggingface.co/deepseek-ai/Janus-Pro-1B (Janus-Pro-1B)

Fine-tuning method:

- **LoRA** (recommended)

It is not a must to use unified MLLM, you can also use Diffusion-based or Flow-based t2i models or frameworks like FlowGRPO. Lora is not a must as well, as long as you can fit in the memory.

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4. Pipeline Overview

Prompt

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Image Generator

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VLM / Reward Model (you can call API-based VLM model as the online memory is low)

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Scalar Reward

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RL or Reward-Weighted Update

Reward signals:

- CLIP similarity,
- VLM-based object / attribute correctness
- checklist-based evaluation.

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5. Evaluation Requirements

Report:

- benchmark scores (choose ≥3 from above or other ones that applied),
- qualitative before/after grids,
- error taxonomy (missing, wrong count, wrong attribute, wrong relation).

Deliverables (All Topics)

1. **Code**
    - training scripts,
    - evaluation scripts,
    - reproducible README.
2. **Final Report (6–8 pages)**
    - background + related work,
    - method & algorithm overview,
    - experiments,
    - analysis & limitations.
3. **Presentation (specific info to be announced later)**

Project Takeaways

After completing this project, you should be able to:

- design an end-to-end training + evaluation pipeline,
- understand why alignment methods work or fail,
- reason about failure modes beyond leaderboard scores,
- operate under realistic compute constraints.

Flexibility & Grading Policy (Important)

The pipeline stages, analysis items, suggested base models, training frameworks, and benchmark list above are **recommendations for reference**, not strict requirements. You are **free to modify** the pipeline design (e.g., different data selection strategy, SFT/RL framework, reward definition, model choice, or evaluation suite) as long as your choices are **well-motivated**, **clearly documented**, and lead to **meaningful experimental evidence**.

We will evaluate your project primarily by **overall quality**, including:

- clarity of problem setup and motivation,
- soundness of methodology and reward design (for RL),
- rigor and completeness of experiments,
- quality of analysis (ablations, failure modes, trade-offs),
- reproducibility (clean code, configs, clear reporting),

not by strict adherence to the exact models / datasets / tools / benchmarks listed above.